King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Major Exam I Term 223

Original exam date: 12 July 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. If the graph of the function $f(x) = \sqrt{3}\,x + 2\cos x$, $0 \le x < 2\pi$ has a horizontal tangent line at $x = a$, then the sum of all a's is

(a) π

(b) 0

(c) 4π

(d) 2π

(e) 3π

2. $\lim_{x\to 1} \frac{\ln x}{x-1} + \lim_{x\to \ln 2} \frac{e^{3x}-8}{e^{2x}-4} =$

(a) 3

(b) 5

(c) 4

(d) 6

(e) 2

3. The product of all k's for which $y = k$ is a horizontal asymptote for the function $f(x) = \dfrac{|x|}{x+1}$ is

(a) -1

(b) -2

(c) 2

(d) 1

(e) 0

4. If the line $y = x + 4$ is tangent to the graph of the function $f(x) = k\sqrt{x}$, then $k =$

(a) -2

(b) 0

(c) 4

(d) -4

(e) 2

5. $\lim_{x\to 1} (\lfloor x \rfloor + \lfloor -x \rfloor) =$

(a) 0

(b) 1

(c) -1

(d) does not exist

(e) 2

6. $\lim_{x\to 0} \frac{\sec x - 1}{x^2} =$

(a) 1

(b) 0

(c) -1

(d) $\frac{1}{2}$

(e) $-\frac{1}{2}$

7. If

$$f(x) = \begin{cases} \cos x, & x < 0 \\ ax + b, & x \geq 0 \end{cases}$$

is differentiable everywhere, then $a + b =$

(a) -2

(b) 1

(c) 0

(d) 2

(e) -1

8. If the tangent line to the graph of $y = g(x)$ at the point $(4, 5)$ passes through the point $(7, 0)$, then $g(4) + g'(4) =$

(a) $\frac{7}{3}$

(b) $\frac{10}{3}$

(c) 0

(d) 12

(e) 5

9. If $f(x) = (2x^3 + 5x)(x - 3)(x + 2)$, then $f'(0) =$

(a) -20

(b) -50

(c) -40

(d) -30

(e) -10

10. If $y = -e^x + \tan x$, then $y'(0) =$

(a) -2

(b) -1

(c) 0

(d) 2

(e) 1

11. If

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x}, & x \neq 0 \\ 0, & x = 0, \end{cases}$$

then $f'(0)$

(a) equals 1

(b) equals π

(c) equals zero

(d) does not exist

(e) equals -1

12. Let

$$f(x) = \begin{cases} x^2, & x \leq 2 \\ 8 - 2x, & 2 < x < 4 \\ 4, & x \geq 4 \end{cases}$$

The sum of all values of c for which $\lim_{x \to c} f(x)$ does not exist is

(a) -4

(b) 4

(c) 6

(d) 0

(e) 2

13. The function $f(x) = 2e^{\lfloor x \rfloor/4}$ has a nonremovable discontinuity at $x =$

(a) $\frac{1}{2}$

(b) 1

(c) $-\frac{1}{2}$

(d) $-\frac{3}{2}$

(e) $\frac{3}{2}$

14. If the function

$$f(x) = \begin{cases} ae^{x-1} + 3, & x < 1 \\ \tan^{-1}(x-1) + 2, & x \geq 1 \end{cases}$$

is continuous on the entire real number line, then $a =$

(a) 2

(b) 0

(c) -1

(d) -2

(e) 1

15. If $f(x) = \dfrac{e^x}{x+4}$, then the equation of the tangent line to the graph of f at the point $\left(0, \frac{1}{4}\right)$ is $y =$

(a) $-\frac{1}{16}x + \frac{1}{4}$

(b) $\frac{5}{16}x + \frac{1}{4}$

(c) $\frac{3}{16}x + \frac{1}{4}$

(d) $\frac{7}{16}x + \frac{1}{4}$

(e) $\frac{1}{16}x + \frac{1}{4}$

16. If $f''(x) = -\sin x$, then $f^{(8)}\left(\frac{\pi}{2}\right) =$

(a) π

(b) $-\pi$

(c) -1

(d) 1

(e) 0

17. The sum of all k's for which $x = k$ is a vertical asymptote for the function $f(x) = \sec \dfrac{\pi x}{2}$ on the interval $(0, 6)$ is

(a) 7

(b) 9

(c) 8

(d) 11

(e) 10

18. $\lim_{x \to -\infty} \left(3x + \sqrt{9x^2 - x}\right) =$

(a) $\dfrac{1}{3}$

(b) $\dfrac{1}{6}$

(c) $\dfrac{1}{4}$

(d) $-\dfrac{1}{6}$

(e) $-\dfrac{1}{3}$

Answer Key

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Q	Ans	Brief explanation
1	**(a)**	$f' = \sqrt{3} - 2\sin x = 0 \Rightarrow x = \frac{\pi}{3}, \frac{2\pi}{3}$; sum $= \pi$.
2	**(c)**	First limit $= 1$; with $u = e^x$ the second is $\frac{u^3-8}{u^2-4} \to 3$. Sum $= 4$.
3	**(a)**	$\lim_{x\to\infty} = 1$, $\lim_{x\to-\infty} = -1$; product $= -1$.
4	**(c)**	$\frac{k}{2\sqrt{a}} = 1$, $k\sqrt{a} = a + 4 \Rightarrow k^2 = 16$; $\sqrt{a} = k/2 > 0$ gives $k = 4$.
5	**(c)**	Left: $0 + (-1) = -1$. Right: $1 + (-2) = -1$. Limit $= -1$.
6	**(d)**	$\frac{\sec x - 1}{x^2} = \frac{\sin^2 x}{x^2} \cdot \frac{1}{\cos x(1+\cos x)} \to \frac{1}{2}$.
7	**(b)**	Continuity: $b = \cos 0 = 1$; slopes: $-\sin 0 = a$ gives $a = 0$. So $a + b = 1$.
8	**(b)**	$g(4) = 5$, $g'(4) = \frac{0-5}{7-4} = -\frac{5}{3}$, so the sum is $\frac{10}{3}$.
9	**(d)**	Since $u(0) = 0$: $f'(0) = u'(0)\,v(0)\,w(0) = 5 \cdot (-3) \cdot 2 = -30$.
10	**(c)**	$y' = -e^x + \sec^2 x$, so $y'(0) = -1 + 1 = 0$.
11	**(c)**	$f'(0) = \lim_{h\to 0} h \sin\frac{1}{h} = 0$ by the Squeeze Theorem.
12	**(b)**	Only $c = 4$ fails (left limit 0, right limit 4); $c = 2$ gives $4 = 4$. Sum $= 4$.
13	**(b)**	$\lfloor x \rfloor$ jumps at integers, so f has jump discontinuities there; only $x = 1$ is an integer.
14	**(c)**	Match at $x = 1$: $a + 3 = \tan^{-1}(0) + 2 = 2$, so $a = -1$.
15	**(c)**	$f'(x) = \frac{e^x(x+3)}{(x+4)^2}$, $f'(0) = \frac{3}{16}$: $y = \frac{3}{16}x + \frac{1}{4}$.
16	**(d)**	Derivatives cycle with period 4, so $f^{(8)} = \sin x$ and $f^{(8)}(\pi/2) = 1$.
17	**(b)**	$\cos\frac{\pi x}{2} = 0$ at odd x; in $(0, 6)$: $1 + 3 + 5 = 9$.
18	**(b)**	Conjugate gives $\frac{-x}{\sqrt{9x^2-x}-3x} = \frac{1}{\sqrt{9-1/x}+3} \to \frac{1}{6}$.